Third Implementation Iteration: One-Time Space Allocation

with Classroom and Laboratory Resource Types

Software Design / Iteration Report

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1. Iteration Goal

The goal of this third implementation iteration is to extend the resource allocation prototype by adding a second resource type. In the previous iteration, the system supported only classroom allocation. In this iteration, the system is generalized to support both `Classroom` and `Laboratory` resources.

To achieve this, a more general abstraction, `Space`, is introduced as a common base type for space resources. The request and allocation logic are updated so that the same allocation process can operate on multiple space types while preserving the existing rules: `AvailabilityRule` and `CapacityRule`.

2. Changes from Iteration 2

Compared with the second iteration, the following changes were introduced:

- Introduced a general resource class: `Space`
- Updated `Classroom` to inherit from `Space`
- Added a new resource type: `Laboratory`
- Generalized the request and allocation process from classroom-specific allocation to space allocation
- Preserved the same decision rules:
  - `AvailabilityRule`
  - `CapacityRule`

3. Scenario

A user submits a one-time request for a specific space resource, which may be either a classroom or a laboratory, for a given time slot and a specified number of participants. The system checks whether the requested space is available during that time slot and whether the capacity of the space is sufficient for the participant count.

If both conditions are satisfied, an allocation is created. Otherwise, the request is rejected.

4. Modules and Their Tasks

A. Request Management Module

- Receive the one-time space request from the user
- Create and store `OneTimeRequest`
- Include request details:
  - requester
  - requested space
  - requested time slot
  - participant count
- Send the request to the Rule Evaluation Module

B. Resource Management Module

- Manage multiple space resource types
- Provide resource details for `Classroom` and `Laboratory`
- Provide space properties such as identity, name, type, and capacity

C. Rule Evaluation Module

- Evaluate the request using `AvailabilityRule`
- Evaluate the request using `CapacityRule`
- Check whether the requested space is already allocated during the requested time slot
- Check whether the space capacity is sufficient for the participant count
- Return the final decision: Approved or Rejected

D. Allocation Management Module

- Create `Allocation` if all rules pass
- Store the new allocation
- Update request status to Approved
- Reject the request if one or more rules fail
- Update request status to Rejected

5. Iteration Comparison

Iteration	Request Type	Resource Types	Rules Implemented
Iteration 1	OneTimeRequest	Classroom	AvailabilityRule
Iteration 2	OneTimeRequest	Classroom	AvailabilityRule + CapacityRule
Iteration 3	OneTimeRequest	Classroom + Laboratory	AvailabilityRule + CapacityRule

6. Updated Class Diagram

Figure 1 shows the updated simplified class diagram for the third implementation iteration.

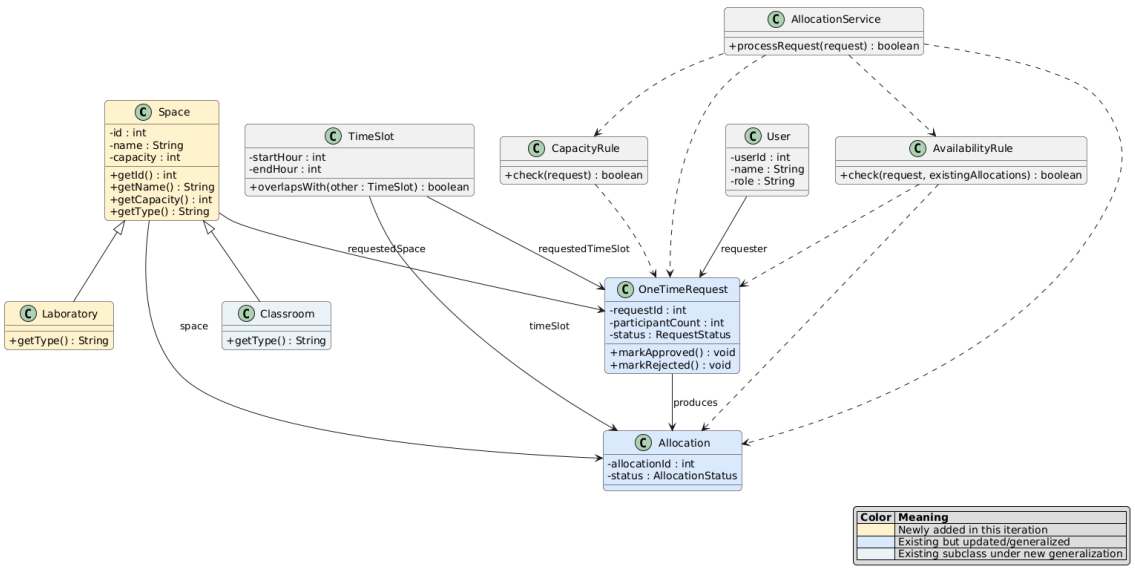

Figure 1: Updated Simplified Class Diagram

7. Updated Workflow Diagram

Figure 2 shows the workflow for the third iteration of the one-time space allocation process.

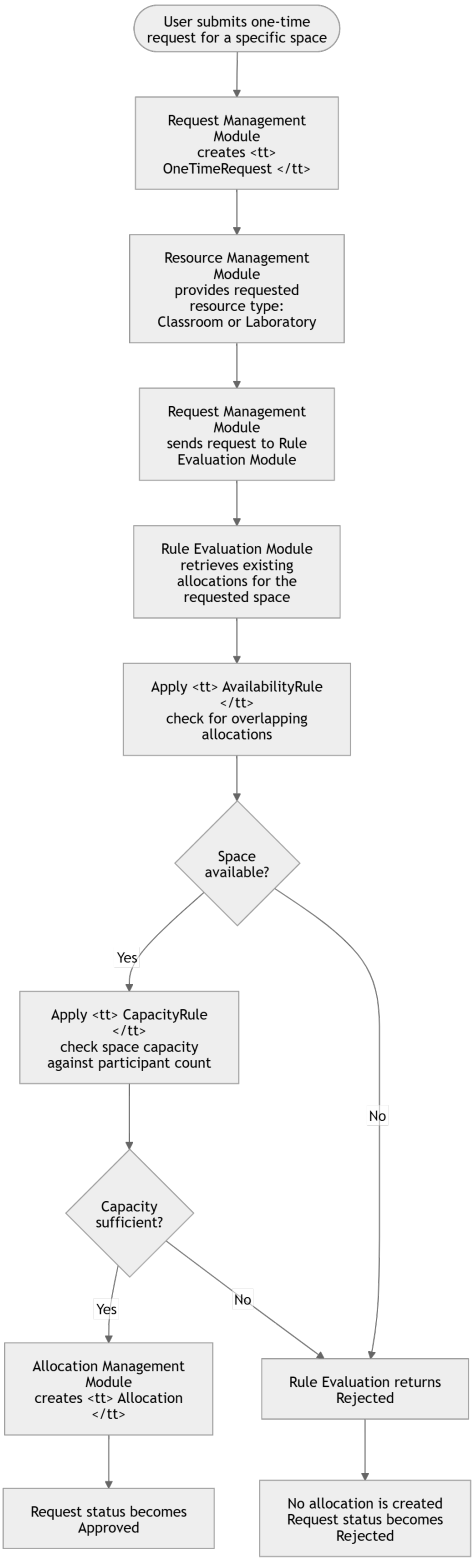

Figure 2: Workflow Diagram

8. Prototype Implementation

To validate the design of this iteration, the C++ prototype was refactored to support multiple space resource types. A new abstract resource concept, `Space`, was introduced, and the existing classroom logic was generalized so that the same request-processing mechanism can be used for both classrooms and laboratories.

The prototype was tested using multiple scenarios involving both resource types. These scenarios confirm that the system can correctly evaluate availability and capacity for both classrooms and laboratories using the same allocation flow.

9. Summary

This third implementation iteration extends the prototype by adding support for a second resource type while preserving the existing request type and rules. The system now processes one-time requests for both classrooms and laboratories.

This iteration demonstrates the iterative development approach by generalizing the resource model without changing the overall decision mechanism. Future iterations can extend this structure further by introducing additional space types, more request types, or additional allocation rules.